PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Jackfruit Phase-1

Software Requirements Specification

For **LinkedIn and Social Media–Driven Automated Job Application System**

Version 1.0 approved

Team details

Team Name	Project ID selected	Team Member 1 Name and SRN	Team Member 2 Name and SRN	Team Member 3 Name and SRN	Team Member 4 Name and SRN
HAND	16	**H**uma PES2UG24AM066	**A**shmitha Sri Anand PES2UG24AM031	Chennupati **N**iraj Kumar PES2UG24AM046	**D**hruv Jagadeesh Hemmige PES2UG24AM053

Prepared by: Team HAND

PES University, Department of Computer Science and Engineering (AI/ML), EC Campus

Date created: 05 September 2026

Table of Contents

Table of Contents 2
Revision History 3
1. Introduction 4
1.1 Purpose 4
1.2 Intended Audience and Reading Suggestions 4
1.3 Product Scope 4
1.4 References 4
2. Overall Description 5
2.1 Product Perspective 5
2.2 Product Functions 5
2.3 User Classes and Characteristics 5
2.4 Operating Environment 5
2.5 Design and Implementation Constraints 6
2.6 Assumptions and Dependencies 6
3. External Interface Requirements 7
3.1 User Interfaces 7
3.2 Software Interfaces 7
3.3 Communications Interfaces 7
4. Analysis Models 8
5. System Features 9
5.1 Resume Parsing and Skill-Profile Extraction 9
5.2 Multi-Platform Job Search and Relevance Matching 9
5.3 Automated Application Submission 10
5.4 Recruiter Networking and Outreach 10
5.5 Scheduling and Rate-Limit Compliance 11
5.6 Application Tracking and Analytics Dashboard 11
6. Other Nonfunctional Requirements 12
6.1 Performance Requirements 12
6.2 Safety Requirements 12
6.3 Security Requirements 12
6.4 Software Quality Attributes 12
6.5 Business Rules 12
7. Other Requirements 14
Appendix A: Glossary 15
Appendix B: Field Layouts 16
Appendix C: Requirement Traceability Matrix 17

Revision History

Name	Date	Reason For Changes	Version
Team HAND	05-Sep-2026	Initial draft of SRS for Jackfruit Phase-1 submission	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for Version 1.0 of the "LinkedIn and Social Media–Driven Automated Job Application System" (internally referred to as HAND-JobConnect), being developed by Team HAND (Project ID 16) as part of the Jackfruit Software Engineering Mini-Project (Subject Code UE24CS341A) at PES University. This SRS covers the complete scope of the system, including resume-driven job discovery, automated applications, recruiter networking, and analytics across LinkedIn, X (Twitter), and Naukri.com. It does not cover infrastructure provisioning or the university's evaluation/grading workflow, which are outside the scope of the product itself.

1.2 Intended Audience and Reading Suggestions

This document is intended for the four members of Team HAND (developers and testers for the project), the course instructor and evaluators (Dr. Siba Ram Baral and panel) who will review the requirements for completeness and traceability, and any future maintainers of the codebase. Section 2 gives a high-level overview suited to first-time readers; Section 3 and Section 5 give detailed, testable requirements meant for developers and testers; Section 6 and the Appendices are meant for reviewers checking non-functional coverage and traceability.

1.3 Product Scope

Job hunting for students and early-career professionals is fragmented across multiple platforms (LinkedIn, X/Twitter, Naukri.com and others), each with its own search interface, application form, and networking mechanism. HAND-JobConnect aims to reduce this manual overhead by parsing a user's resume once, deriving a structured skill/role/location profile from it, and using that profile to discover, rank, and (with explicit user consent) apply to matching postings across the three supported platforms from a single dashboard. The product also assists users in professional networking on LinkedIn/X by suggesting and sending personalized connection requests to relevant recruiters, and gives users visibility into their application pipeline through a unified analytics dashboard. The system is built to respect each platform's terms of service and rate limits, and always keeps the user informed and in control (no submission happens without an explicit review step, unless the user opts into a fully-automated mode for a given platform).

1.4 References

- IEEE SRS Template, as provided in "JACKFRUIT -SE MINI-PROJECT GUIDELINES 2026" document, PES University, Dept. of CSE.
- LinkedIn API / Marketing Developer Platform documentation – https://learn.microsoft.com/en-us/linkedin/
- X (Twitter) API v2 documentation – https://developer.twitter.com/en/docs/twitter-api
- Naukri.com public site and RSS/search endpoints (used for scraping/automation reference).
- Selenium WebDriver documentation – https://www.selenium.dev/documentation/
- Django REST Framework documentation – https://www.django-rest-framework.org/

2. Overall Description

2.1 Product Perspective

HAND-JobConnect is a new, self-contained web application; it is not a replacement for an existing system. It sits between the user and three external platforms (LinkedIn, X/Twitter, Naukri.com), acting as an orchestration layer that combines official APIs (where available, e.g., X API v2) with browser-automation (Selenium) for platforms that do not expose a public application API (e.g., LinkedIn Easy Apply, Naukri.com). At a high level the system comprises: a Resume Parsing module, a Job Aggregation & Matching engine, an Automated Application module, a Recruiter Networking module, and an Analytics Dashboard, all backed by a central Django application server and a relational database.

2.2 Product Functions

- Parse an uploaded resume (PDF/DOCX) and extract skills, education, experience, and preferred roles/locations.
- Search and aggregate job postings from LinkedIn, X/Twitter (job-related posts/threads) and Naukri.com based on the extracted profile.
- Rank and filter aggregated postings using keyword-weighted relevance scoring.
- Auto-fill and submit job applications on supported platforms, with a mandatory user review step before submission (configurable to fully automatic per platform).
- Identify and send personalized connection/outreach requests to recruiters on LinkedIn and X.
- Schedule recurring search-and-apply runs and enforce per-platform rate limits.
- Track the status of every application (applied, viewed, rejected, interview) and present it on an analytics dashboard.
- Notify the user of new matches, scheduled-run results, and status changes via in-app notifications and email.

2.3 User Classes and Characteristics

Job Seeker (primary user): Students or early-career professionals who upload a resume, configure search preferences, and review/approve applications. Assumed to have basic web literacy but no technical background. This is the most important user class.

System Administrator: Responsible for managing platform API credentials, monitoring scheduled jobs, viewing system-wide logs, and handling CAPTCHAs/login failures escalated by the automation engine. Assumed to have technical proficiency.

Guest/Unregistered Visitor: Can view the landing page and product information but cannot upload a resume or run searches until registered. Least important user class from a requirements standpoint.

2.4 Operating Environment

- Server: Python 3.11+, Django 5.x with Django REST Framework, deployed on a Linux (Ubuntu 22.04 LTS) host.
- Database: PostgreSQL 15.
- Automation: Selenium 4 WebDriver with headless Chromium, run inside Docker containers for isolation.
- Client: Any modern evergreen browser (Chrome, Firefox, Edge, Safari) on desktop or mobile, minimum resolution 360px width.
- Task Scheduling: Celery with Redis as the message broker, used for scheduled search-and-apply runs.

2.5 Design and Implementation Constraints

- The system must comply with the Terms of Service and rate limits of LinkedIn, X, and Naukri.com; excessive or non-compliant automation (e.g., mass unsolicited messaging) is explicitly out of scope and disallowed by design.
- LinkedIn and Naukri.com do not offer a public application-submission API, so those integrations depend on Selenium browser automation, which is inherently more fragile to UI changes than API calls.
- X (Twitter) API v2 access is rate-limited per the developer tier used by the team (Free/Basic tier assumed for this project), which limits the volume of searches per 15-minute window.
- All credentials (platform login/API keys) must be stored encrypted at rest; the system must never store a user's plaintext password for a third-party platform beyond the single automation session.
- The project must follow Agile methodology with GitHub (Issues/Projects) used for backlog and sprint tracking, and CI/CD (GitHub Actions) with mandatory code review before merge, per the Jackfruit guidelines.

2.6 Assumptions and Dependencies

- It is assumed that each user maintains an active, individually-owned account on LinkedIn, X, and/or Naukri.com; the system does not create third-party accounts on the user's behalf.
- It is assumed that resumes are supplied in a parseable text-based PDF or DOCX format; scanned/image-only resumes are out of scope for automatic parsing.
- The project depends on continued availability of the X API v2 free/basic tier and on LinkedIn/Naukri.com not introducing anti-automation measures (e.g., CAPTCHAs) that block Selenium sessions; if such measures appear, manual-review fallback is used.
- The project depends on third-party libraries: pdfplumber/python-docx (resume parsing), spaCy (skill/entity extraction), Selenium, and Tweepy (X API client).

3. External Interface Requirements

3.1 User Interfaces

The system exposes a responsive web UI built with a component-based front end (React) consuming the Django REST Framework API. Key screens include: Registration/Login, Resume Upload & Profile Review, Search Preferences configuration, Job Feed (aggregated, ranked postings with Apply/Skip actions), Application Review modal (shown before any submission unless full-auto is enabled), Recruiter Outreach panel, and the Analytics Dashboard. All screens share a common header/navigation bar, a consistent primary-action button style, and standard toast-style error/success messages. Destructive or platform-facing actions (e.g., "Send Application", "Send Connection Request") always show a confirmation step. Keyboard navigation and WCAG 2.1 AA color-contrast are followed for accessibility.

3.2 Software Interfaces

- LinkedIn: No public application-submission API is used; browser automation (Selenium) drives the "Easy Apply" flow and connection-request flow using the user's authenticated session. Read-only job search uses LinkedIn's public job-search pages.
- X (Twitter) API v2: Used via the Tweepy client to search job-related tweets/threads (filtered by hashtags/keywords such as #hiring, #jobs) and to send direct messages/replies for outreach, subject to the developer tier's rate limits.
- Naukri.com: Accessed via Selenium browser automation for search, filtering, and application submission, since no public API is offered.
- Database: PostgreSQL 15, accessed through Django ORM; stores users, resumes (parsed profile only – not raw file after parsing, unless the user opts to retain it), job postings, applications, and connection requests.
- Email service: Integrates with an SMTP provider (e.g., SendGrid) for notification emails (new matches, weekly summary, failures requiring manual review).

3.3 Communications Interfaces

All communication between the client and the Django REST API uses HTTPS (TLS 1.2+). REST endpoints exchange JSON payloads. Scheduled background jobs communicate with the web server via Redis-backed Celery task queues over an internal network, not exposed externally. Session authentication uses JWT access/refresh tokens; platform credentials used by the Selenium automation layer are passed over an internal encrypted channel and never logged in plaintext.

4. Analysis Models

The following high-level data flow describes the core use case ("Search and Apply"): (1) User uploads resume → Resume Parser extracts a structured Profile; (2) User confirms/edits Search Preferences (roles, locations, platforms); (3) Job Aggregation & Matching engine queries LinkedIn, X, and Naukri.com connectors and produces a ranked Job Feed; (4) User reviews and selects postings (or enables full-auto for a platform); (5) Automated Application module submits applications via the relevant connector and records an Application entity; (6) Analytics Dashboard aggregates Application status over time. A detailed use-case diagram (actors: Job Seeker, System Administrator; use cases: Upload Resume, Configure Preferences, Review Job Feed, Apply to Job, Send Connection Request, Schedule Recurring Search, View Analytics) and an entity-relationship diagram (entities: User, Resume, Profile, JobPosting, Application, ConnectionRequest, ScheduledRun) will be maintained in the project's GitHub repository under /docs/diagrams and referenced in the Phase-2 Project Plan and Design documents.

5. System Features

5.1 Resume Parsing and Skill-Profile Extraction

5.1.1 Description and Priority

Resume Parsing and Skill-Profile Extraction is owned primarily by Huma (PES2UG24AM066). Priority: High.

5.1.2 Stimulus/Response Sequences

- User uploads a resume file → system extracts text, identifies skills/education/experience/preferred roles, and displays an editable Profile summary.
- User edits an incorrectly-extracted field → system saves the correction and uses it for all future matching.

5.1.3 Functional Requirements

REQ-1: The system shall accept resume uploads in PDF and DOCX format up to 5 MB.

REQ-2: The system shall extract, at minimum, the candidate's skills, total years of experience, highest qualification, and up to five preferred job titles from the uploaded resume.

REQ-3: The system shall present the extracted Profile to the user for review and allow manual editing of every extracted field before it is used for job matching.

REQ-4: The system shall reject uploads that are not parseable as text (e.g., scanned image-only PDFs) and shall prompt the user to provide a text-based resume.

5.2 Multi-Platform Job Search and Relevance Matching

5.2.1 Description and Priority

Multi-Platform Job Search and Relevance Matching is owned primarily by Dhruv Jagadeesh Hemmige (PES2UG24AM053). Priority: High.

5.2.2 Stimulus/Response Sequences

- User confirms search preferences (roles, locations, platforms, experience level) → system queries LinkedIn, X, and Naukri.com connectors and returns a ranked Job Feed within 30 seconds for a manual search.
- A scheduled run executes automatically → system refreshes the Job Feed with newly-found postings and flags duplicates already seen.

5.2.3 Functional Requirements

REQ-5: The system shall allow the user to select one or more of LinkedIn, X (Twitter), and Naukri.com as active search sources.

REQ-6: The system shall compute a relevance score for each job posting based on weighted keyword overlap between the posting and the user's Profile, and shall sort the Job Feed by descending relevance score.

REQ-7: The system shall detect and suppress duplicate postings for the same role/company appearing across multiple platforms or multiple search runs.

REQ-8: The system shall allow the user to filter the Job Feed by platform, location, minimum relevance score, and posting recency.

5.3 Automated Application Submission

5.3.1 Description and Priority

Automated Application Submission is owned primarily by Chennupati Niraj Kumar (PES2UG24AM046). Priority: High.

5.3.2 Stimulus/Response Sequences

- User selects a posting and clicks "Apply" → system opens an Application Review modal showing the auto-filled form fields → user confirms → system submits the application via the relevant connector and records the result.
- User enables "Full-Auto" mode for a platform → system applies to postings above a user-defined relevance threshold without per-application confirmation, subject to a daily application cap.

5.3.3 Functional Requirements

REQ-9: The system shall auto-fill the platform-specific application form using data from the user's Profile and resume.

REQ-10: The system shall require explicit user confirmation before submitting an application unless the user has enabled Full-Auto mode for that platform.

REQ-11: The system shall enforce a configurable daily application cap per platform (default: 20 applications/day) to remain within each platform's acceptable-use limits.

REQ-12: The system shall record every submission attempt (success, failure, or CAPTCHA/manual-review-required) as an Application entity with a timestamp and status.

REQ-13: If a CAPTCHA or login challenge is encountered during automated submission, the system shall pause the automation session, notify the user/administrator, and mark the application as "Manual Review Required" rather than failing silently.

5.4 Recruiter Networking and Outreach

5.4.1 Description and Priority

Recruiter Networking and Outreach is owned primarily by Chennupati Niraj Kumar (PES2UG24AM046). Priority: Medium.

5.4.2 Stimulus/Response Sequences

- System identifies a recruiter profile associated with a matched posting on LinkedIn/X → user reviews a suggested, personalized connection message → user approves → system sends the connection/outreach request.

5.4.3 Functional Requirements

REQ-14: The system shall suggest recruiter or hiring-manager profiles associated with a job posting where such association is discoverable via the platform.

REQ-15: The system shall generate a personalized outreach message template referencing the specific posting and the user's relevant skills, and shall allow the user to edit the message before sending.

REQ-16: The system shall enforce a configurable daily connection-request cap per platform (default: 15/day on LinkedIn, per LinkedIn's informal invitation-abuse thresholds) to avoid account restriction.

REQ-17: The system shall never send a bulk/identical message to more than one recipient without per-recipient personalization.

5.5 Scheduling and Rate-Limit Compliance

5.5.1 Description and Priority

Scheduling and Rate-Limit Compliance is owned primarily by Dhruv Jagadeesh Hemmige (PES2UG24AM053). Priority: Medium.

5.5.2 Stimulus/Response Sequences

- User configures a recurring schedule (e.g., daily at 9 AM) → system runs the search-and-apply pipeline automatically at the configured time and logs the run.

5.5.3 Functional Requirements

REQ-18: The system shall allow the user to schedule recurring search runs at a daily or weekly cadence.

REQ-19: The system shall track, per platform, the number of API calls/automation actions performed within the platform's rate-limit window and shall defer further actions once 90% of the limit is reached.

REQ-20: The system shall log the outcome (postings found, applications submitted, errors) of every scheduled run and make the log available to the user.

5.6 Application Tracking and Analytics Dashboard

5.6.1 Description and Priority

Application Tracking and Analytics Dashboard is owned primarily by Ashmitha Sri Anand (PES2UG24AM031). Priority: High.

5.6.2 Stimulus/Response Sequences

- User opens the Analytics Dashboard → system displays counts and trends of applications by status (Applied, Viewed, Interview, Rejected, Offer) and by platform.
- An application's status changes (detected via connector polling or user manual update) → dashboard updates in near real time and, if configured, sends a notification.

5.6.3 Functional Requirements

REQ-21: The system shall display, for a selectable date range, the count of applications submitted, broken down by platform and by status.

REQ-22: The system shall allow the user to manually update the status of an application (e.g., mark as "Interview Scheduled") when the platform does not expose status updates programmatically.

REQ-23: The system shall compute and display a response rate (interviews or replies ÷ applications submitted) overall and per platform.

REQ-24: The system shall allow the user to export their application history as a CSV file.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- A manual job search across all three enabled platforms shall return an initial ranked Job Feed within 30 seconds under normal network conditions.
- The Analytics Dashboard shall load and render summary charts within 3 seconds for a user with up to 2,000 recorded applications.
- The system shall support at least 50 concurrent users without a response-time degradation of more than 20% relative to single-user baseline, for the purposes of this academic project's load testing.

6.2 Safety Requirements

- The system shall never submit an application or send an outreach message without the safeguards defined in REQ-10 and REQ-15 (explicit confirmation, unless Full-Auto is knowingly enabled).
- The system shall provide a visible "Pause All Automation" control that immediately halts all scheduled runs and in-flight automation sessions.

6.3 Security Requirements

- The system shall store all third-party platform credentials and API tokens encrypted at rest (AES-256) and shall never write plaintext credentials to logs.
- The system shall authenticate users via hashed (bcrypt/argon2) passwords and JWT-based session tokens with a maximum access-token lifetime of 30 minutes.
- The system shall enforce role-based access control distinguishing Job Seeker and System Administrator capabilities, per Section 6.5.
- All external API/browser-automation traffic shall occur over HTTPS/TLS.

6.4 Software Quality Attributes

- Maintainability: The codebase shall follow a modular architecture (separate packages for resume-parsing, connectors, scheduling, and analytics) to allow a platform connector to be added or modified without changes to unrelated modules.
- Reliability: Automated application submissions shall achieve at least 90% successful-submission rate (excluding CAPTCHA/manual-review cases) across supported platforms, measured during system testing.
- Usability: A first-time user shall be able to upload a resume and reach a populated Job Feed within 3 user actions/screens.
- Testability: Each functional requirement in Section 5 shall map to at least one automated or manual test case, per the Requirement Traceability Matrix in Appendix C.

6.5 Business Rules

- Only an authenticated Job Seeker may upload a resume, configure preferences, or trigger/approve applications for their own account.
- Only a System Administrator may view system-wide automation logs, manage platform API credentials, or override a paused automation session.
- A Job Seeker may not exceed the configured daily application or connection-request caps defined in REQ-11 and REQ-16, regardless of Full-Auto settings.

- Domain requirement: the system must remain compliant with each connected platform's Terms of Service; if a platform is detected to have changed its terms in a way that prohibits automated applications, that platform's connector shall be disabled pending review.

7. Other Requirements

- Database requirement: all Profile, JobPosting, Application, and ConnectionRequest data shall be retained for the duration of the user's active account and purged within 30 days of account deletion, in line with data-minimization practice.
- Internationalization: the initial release targets English-language resumes and postings only; multi-language support is a future enhancement and out of scope for Version 1.0.
- Legal requirement: the system shall present users with a clear disclaimer that automated applications are submitted using the user's own platform credentials/consent, and that Team HAND/PES University are not liable for platform account actions (e.g., suspension) resulting from a user exceeding recommended usage caps.
- Reuse objective: the Resume Parsing module is designed to be reusable as a standalone library independent of the platform connectors, for potential reuse in future course projects.

Appendix A: Glossary

Term	Definition
Profile	The structured record of a user's skills, experience, education, and preferences, derived from their resume and manual edits.
Connector	A platform-specific module (LinkedIn, X, Naukri.com) responsible for searching, applying, and/or messaging on that platform.
Easy Apply	LinkedIn's in-platform one-click application feature, driven by the LinkedIn Connector via Selenium automation.
Full-Auto Mode	A per-platform setting that allows the system to submit applications above a relevance threshold without per-application user confirmation.
Relevance Score	A numeric score (0–100) representing the weighted keyword overlap between a job posting and the user's Profile.
Scheduled Run	An automated, recurring execution of the search-and-apply pipeline at a user-configured cadence.
Manual Review Required	An Application status indicating that automation could not complete (e.g., due to a CAPTCHA) and needs user/administrator action.
CI/CD	Continuous Integration / Continuous Deployment; automated build, test, and deployment pipeline configured via GitHub Actions.

Appendix B: Field Layouts

Sample field layout for the Resume Profile record, produced by the Resume Parsing module and consumed by the Job Aggregation & Matching engine:

Field	Length	Data Type	Description	Mandatory
User ID	36	UUID	Unique identifier of the account owning this Profile.	Y
Full Name	100	String	Candidate name as extracted from the resume.	Y
Skills	500	Array<String>	List of extracted technical/soft skills.	Y
Years of Experience	2	Numeric	Total professional experience in years.	Y
Highest Qualification	150	String	e.g., "B.Tech CSE, PES University".	Y
Preferred Roles	300	Array<String>	Up to five preferred job titles.	Y
Preferred Locations	200	Array<String>	Cities/regions the user is targeting.	N
Resume File Reference	260	String (path/URL)	Storage reference to the original uploaded file.	N

Sample report requirements — fields included in the Application History export (CSV):

Application Report	Analytics Summary Report
Application ID	Date Range
Platform	Total Applications
Job Title	Applications by Platform
Company Name	Applications by Status
Relevance Score	Response Rate (%)
Date Applied	Interviews Scheduled
Current Status	Connection Requests Sent
Manual Review Flag	Average Time-to-Response (days)

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID / Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	REQ-1: Accept resume upload (PDF/DOCX, ≤5MB)	Resume Parsing Module	Upload API design	resume_parser/ upload.py	UT-01	ST-01
2	REQ-2: Extract skills/experience/education/roles	Resume Parsing Module	NER pipeline design	resume_parser/ extract.py	UT-02	ST-01
3	REQ-5: Multi-platform search source selection	Job Aggregation Engine	Connector interface design	matching/ search.py	UT-05	ST-02
4	REQ-6: Relevance scoring and ranking	Job Aggregation Engine	Scoring algorithm design	matching/ rank.py	UT-06	ST-02
5	REQ-9: Auto-fill application form	Application Module	Connector form-mapping design	applications/ autofill.py	UT-09	ST-03
6	REQ-10: Confirm before submission (unless Full-Auto)	Application Module	Review-modal design	applications/ submit.py	UT-10	ST-03
7	REQ-13: CAPTCHA/manual-review handling	Application Module	Error-handling design	applications/ errors.py	UT-13	ST-03
8	REQ-14: Suggest recruiter profiles	Networking Module	Recruiter-lookup design	networking/ suggest.py	UT-14	ST-04
9	REQ-18: Recurring scheduled search runs	Scheduling Module	Celery task design	scheduler/ tasks.py	UT-18	ST-05
10	REQ-21: Application analytics by platform/status	Analytics Dashboard	Dashboard query design	analytics/ reports.py	UT-21	ST-06

Note: This matrix will be expanded to cover all 24 functional requirements (REQ-1 through REQ-24) once Architecture and Design references are finalized in Jackfruit Phase-2.